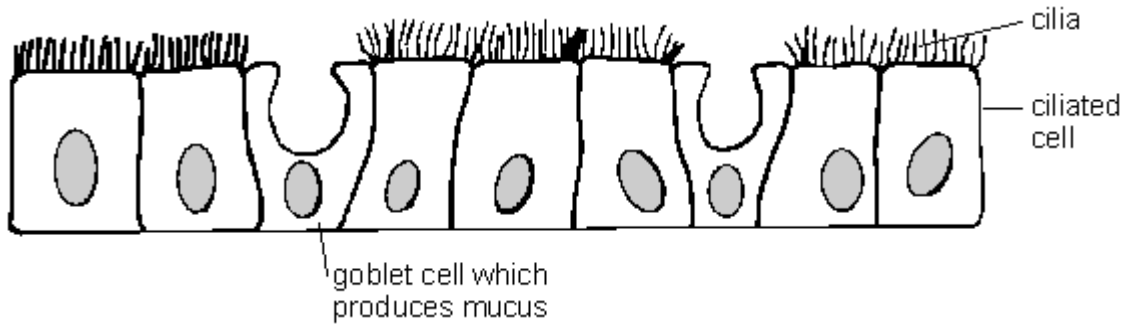
Q1.

The diagram shows two types of cell in the lining of the windpipe.



(a) (i) These cells work together to keep the lungs free of bacteria and dust particles.
What word describes a group of similar cells which work together?

……

1 mark

(ii) Mucus is a sticky substance.
Describe how **mucus** and **cilia** keep the lungs free of bacteria and dust particles.

……

……

……

……

2 marks

(b) When a person breathes in cigarette smoke, the goblet cells produce extra mucus and the cilia are damaged.
What will be the consequences of this?

……

……

……

……

2 marks

(c) Give the names of **two** harmful substances in cigarette smoke.
In what way is each one harmful?

1. name of substance ……

   harmful effect ……..……..

   ……..…….

2. name of substance ……...…….

   harmful effect ……..……

   ……..…….

2 marks
Maximum 7 marks

Q1.

(a) (i) tissue

1 (L7)

(ii) mucus traps bacteria **or** dust

accept 'they are trapped by mucus'

1 (L7)

cilia move mucus out of the lungs **or** windpipe

*accept 'cilia move them upwards' **or** 'cilia move mucus'*

*accept 'cilia remove bacteria **or** dust'*

1 (L7)

(b) any **two** from

- mucus will build up

  accept 'cilia cannot move mucus'

- airways will be partially blocked

  *accept 'blocked airways' **or** named part of the airway*

  *do **not** accept 'lungs will be blocked'*

  accept 'breathing will be more difficult'

- coughing

- lungs can become infected **or** bacteria are not removed

2 (L7)

(c) **Both the name of the substance and its effect are needed for each mark.**

any **two** from

- nicotine causes addiction **or** raised blood pressure **or** raised heart rate **or** stops cilia working

- tar causes cancer **or** blocks the airways **or** clogs the cilia **or** builds up in the alveoli

  *do **not** accept 'tar blocks the lungs'*

- carbon particles cause coughing **or** cancer

  accept 'carbon monoxide reduces the amount of oxygen carried in the blood'

2 (L7)

[7]